

Training Experience

The training process was conducted using the Vision Transformer (ViT-B/16) model fine-tuned on the Sports Classification dataset containing 100 diverse sports categories. Data preparation included automated dataset downloading from Kaggle, structured directory organization, and augmentation techniques such as random rotation, color jittering, and horizontal flipping to improve generalization. The model, pre-trained on ImageNet-1K, was modified by replacing the classification head to match the dataset's class count and trained using the Adam optimizer with a learning rate of $3e-5$ and weight decay of $1e-4$. Training and validation losses were closely monitored through TensorBoardX, ensuring transparent visualization of convergence trends. Over 100 epochs, the model demonstrated stable learning behavior, with the validation accuracy improving from 96.2% in the first epoch to a best performance of 98.8%. The loss consistently decreased while maintaining minimal overfitting. Overall, the experience provided valuable insight into transformer-based transfer learning, optimization strategies, and the effectiveness of visualization tools for performance tracking and model evaluation.

```
%load_ext tensorboard
```

```
%tensorboard --logdir=runs
```



